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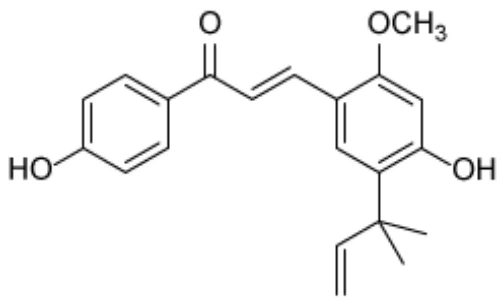
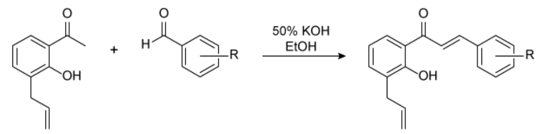


Licochalcone A

Based on these findings, a new panel of chalcone derivatives was prepared as shown in Scheme I. Condensation of 2′-hydroxy-3'-allylacetophenone (Compound **1**) with substituted benzaldehydes (Compound **2**) in a solution of 50% potassium hydroxide in ethanol provided chalcone derivatives (Compound **3**).



Compound **1** Compound **2** Compound **3**

Scheme I

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Experimental data for chalcone derivatives

Compound	R	Molecular formula	Yield (%)	Melting point (°C)	Cell growth (%)
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3f	*p*-NO_2	$C_{18}H_{15}NO_4$	5	169–172	84

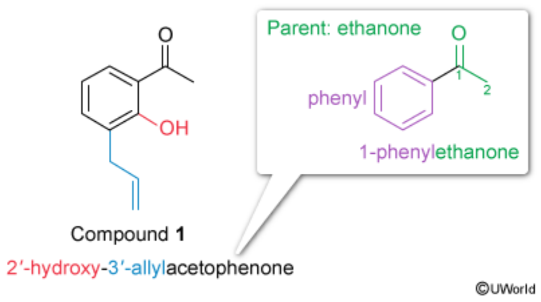
If Compound **1** is a substituted derivative of acetophenone, what is the IUPAC name of acetophenone?

- ○ A. Methyl phenyl ketone [5%]
- ○ B. 1-Benzylethanone [22%]
- ✔ ◉ C. 1-Phenylethanone [57%]
- ○ D. 1-Phenylmethanone [15%]

Correct
56% answered correctly

Explanation:

IUPAC nomenclature of acetophenone



In IUPAC nomenclature, the prime (′) designation usually signifies the substituent position on a ring portion of the molecule rather than the position on the parent chain. In the passage, Compound **1** is identified as 2′-hydroxy-3′-allylacetophenone. An allyl group is the common name for a three-carbon, 2-propenyl substituent, and "hydroxy" refers to the alcohol functional group. Consequently, the acetophenone parent chain must refer to the phenyl ring and the ketone-containing short alkyl chain.

The IUPAC name of a ketone has a parent chain name based on the longest continuous carbon chain that contains the ketone functional group and ends in an *–one* suffix. For acetophenone, the parent chain contains two carbon atoms and is called ethanone. (As there are only two carbon atoms, a numerical location of the ketone group is not necessary.) The **parent chain** is **numbered** to give the ketone functional group the **lowest possible position**. The aromatic benzene substituent is called a phenyl group, and its position on the parent chain is signified by the number "1" before its name. Therefore, the IUPAC name of acetophenone is **1-phenylethanone**.

(Choice A) Methyl phenyl ketone is formatted according to the common guidelines for ketone nomenclature (ie, alkyl or aryl groups are listed alphabetically before the ketone parent name). Therefore, this name does *not* conform to systematic IUPAC guidelines for ketones.

(Choice B) A benzyl group is a substituent containing an aromatic ring and an additional methylene (CH_2) group. This name represents a structure with one too many carbons between the aromatic ring and the carbonyl group.

(Choice D) 1-Phenylmethanone contains the parent chain prefix *meth–*, which refers to structures with only one continuous carbon atom. The parent chain of acetophenone must include both the carbonyl carbon and the adjacent methyl group for a total of two carbons, signified by an *eth–* prefix.

Educational objective:
Ketones are functional groups that contain a carbonyl (C=O) connected to two alkyl or aryl groups. Using IUPAC rules for naming a ketone, the carbon chain containing the ketone is numbered to give the ketone the lowest possible position, and the name will end in the suffix –*one*.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403810
System:
Functional Groups and Their Reactions

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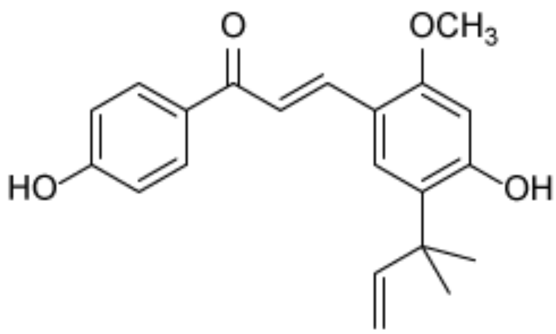
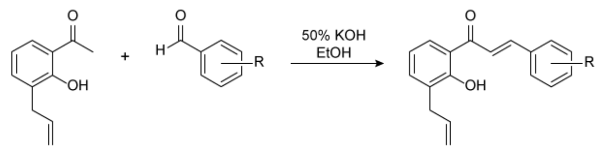


Licochalcone A

Based on these findings, a new panel of chalcone derivatives was prepared as shown in Scheme I. Condensation of 2′-hydroxy-3'-allylacetophenone (Compound **1**) with substituted benzaldehydes (Compound **2**) in a solution of 50% potassium hydroxide in ethanol provided chalcone derivatives (Compound **3**).



Compound **1** Compound **2** Compound **3**

Scheme I

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3f	*p*-NO_2	$C_{18}H_{15}NO_4$	5	169–172	84

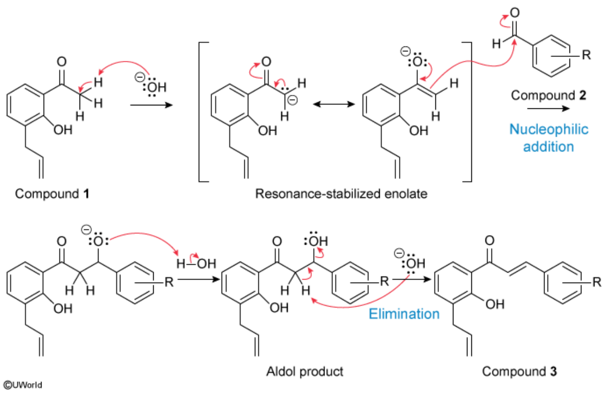
In Scheme I, what is the role of the potassium hydroxide?

- ◯ A. Basic catalyst for a condensation reaction only [38%]
- ◯ B. Basic catalyst for an elimination reaction only [14%]
- ✔ ◉ C. Basic catalyst for both a condensation and an elimination reaction [45%]
- ◯ D. Acidic catalyst for a condensation reaction only [1%]

Correct
45% answered correctly

Explanation:

Base-catalyzed aldol condensation and dehydration



An aldol condensation is an organic transformation between an enolate and a carbonyl-containing ketone or aldehyde. Protons bonded to carbon atoms adjacent to a carbonyl group (an **alpha position**) are much more acidic than isolated hydrocarbons. Deprotonation of alpha protons forms a resonance-stabilized conjugate base called an enolate, which is a strong carbon **nucleophile** at its alpha carbon. The **nucleophilic addition** of an enolate to a carbonyl group in a ketone or an aldehyde generates an alkoxide intermediate. Protonation of the alkoxide intermediate regenerates a molecule of base and forms a β-hydroxycarbonyl (aldol) functional group.

Under strongly basic conditions, the initial aldol product can become further dehydrated to generate an **α,β-unsaturated carbonyl**. A strong base deprotonates a second alpha proton relative to the carbonyl. The resultant electron density forms a new pi bond between the alpha carbon and the beta carbon, and the β-hydroxy functional group becomes the **leaving group** as a molecule of hydroxide.

In Scheme I, Compound **1** and Compound **2** react via an aldol condensation to form Compound **3**. The first step in the mechanism uses the strong base potassium hydroxide to catalyze the condensation of Compounds **1** and **2**, forming the aldol product. Subsequently, the aldol product is dehydrated, eliminating hydroxide and forming the α,β-unsaturated carbonyl product Compound **3**. Therefore, potassium hydroxide is used as a **basic catalyst for both a condensation and an elimination reaction**.

(Choices A and B) Potassium hydroxide serves as a basic catalyst for *both* the condensation and elimination reactions in the aldol condensation mechanism, not just one of these reactions.

(Choice D) The structures of the reactants and products are representative of an aldol condensation with subsequent dehydration. Alkali metal hydroxides are known to be strong bases, which are required for both stages of these synthetic transformations.

Educational objective:
An aldol condensation is the nucleophilic addition of an enolate to either a ketone or an aldehyde. Aldol condensation products often become dehydrated to form an α,β-unsaturated carbonyl product. Both stages of this transformation require a basic catalyst, such as potassium hydroxide.

Time spent:0 sec	**QID**:403811
Subject:	**System:**
Organic Chemistry	Functional Groups and Their Reactions

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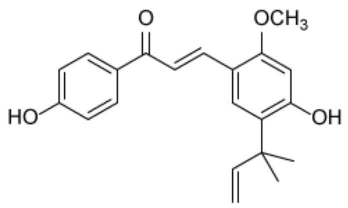

Licochalcone A

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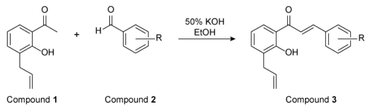

Scheme I

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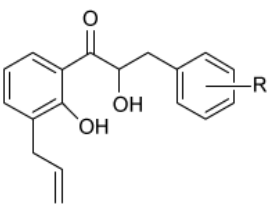
3f	*p*-NO_2	$C_{18}H_{15}NO_4$	5	169–172	84

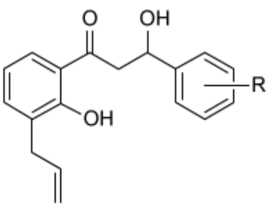
What is the structure of the initial aldol product that is formed in Scheme I?

A.



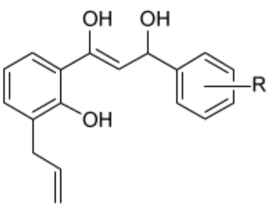
[8%]

✔ B.



[72%]

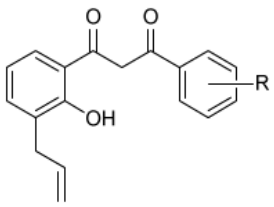
C.



[12%]

D.

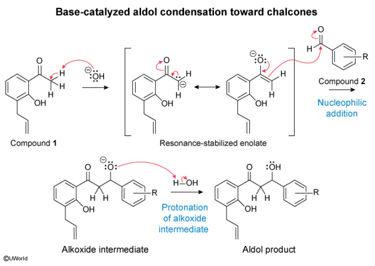
D.



[6%]

Correct
73% answered correctly

Explanation:



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In Scheme I, Compound **1** contains three protons alpha to the carbonyl whereas Compound **2** lacks alpha protons. Therefore, only Compound **1** is capable of undergoing **enolization** in the presence of potassium hydroxide. The following occurs during the **nucleophilic addition** of the enolate of Compound **1** to the aldehyde carbonyl in Compound **2**:

- A new sigma bond is created between the Compound **1** α-carbon and the carbonyl carbon of Compound **2**.
- The ketone carbonyl of Compound **1** is restored.
- The carbonyl pi electrons in Compound **2** become a lone pair on the oxygen atom, forming an alkoxide intermediate.

Protonation of the alkoxide intermediate leads to the structure depicted in Choice B, a β-hydroxyketone (aldol).

(Choice A) The hydroxyl group in this structure is bonded to the alpha carbon of the product scaffold. An α-hydroxycarbonyl is *not* classified as an aldol product.

(Choice C) During the nucleophilic addition stage of an aldol condensation, the carbonyl functional group of the enolate reactant is restored. This structure does not contain a carbonyl group and instead depicts the enol tautomer of the aldol product.

(Choice D) The nucleophilic addition of the Compound **1** enolate to the aldehyde in Compound **2** consumes the carbonyl pi bond from Compound **2** and, following protonation, results in formation of a β-hydroxy group. This structure contains a β-dicarbonyl, which cannot be directly prepared from an aldol condensation.

Educational objective:
An aldol condensation is the nucleophilic addition of a ketone/aldehyde enolate to a second ketone/aldehyde functional group. The mechanism of an aldol condensation demonstrates the location of the new carbon-carbon sigma bond in the aldol (β-hydroxycarbonyl) product.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 403812
System:
Functional Groups and Their Reactions

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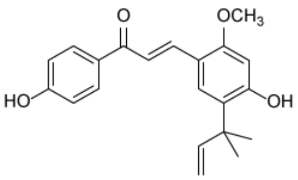

Licochalcone A

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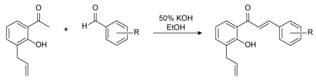

Compound **1** Compound **2** Compound **3**

Scheme I

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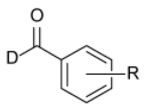
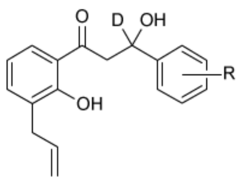
Compound **X** is shown below.



Compound **X**

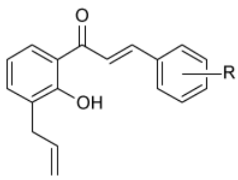
If Compound **2** is replaced with Compound **X**, what will be the structure of the final product of Scheme I? (Note: Assume that the yield of this reaction is not affected by the substitution.)

○ A.



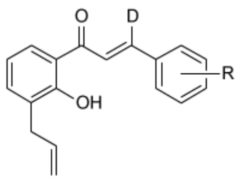
[14%]

○ B.



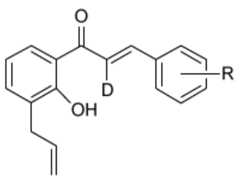
[16%]

✔ ◉ C.



[53%]

○ D.

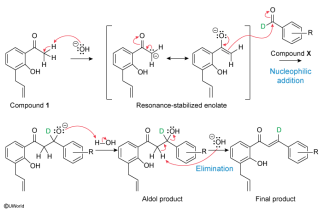


[15%]

Correct
52% answered correctly

Explanation:

Aldol condensation of deuterium-labeled chalcones



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Compound **X** is an aldehyde variant containing a deuterium atom within the aldehyde functional group. Deuterated organic compounds behave similarly to their protonated versions under most circumstances, and the deuterium-containing molecule can be processed through the same aldol reaction mechanism. Based on its location, the deuterium atom position does not play a role in the aldol condensation and dehydration, and the deuterium atom from Compound **X** will be retained through the final product. The deuterium atom will be present at the beta position of the α,β-unsaturated carbonyl product, which corresponds to the structure shown in Choice C.

(Choice A) This structure represents the intermediate aldol product for Scheme I. The inclusion of a deuterium atom in Compound **X** will not impact the dehydration stage of the reaction, and the reaction would not be expected to pause at this intermediate.

(Choice B) This structure represents the original Compound **3** from Scheme I, which would only form if the deuterium was lost through the aldol reaction mechanism. Given the location of the deuterium atom in Compound **X**, the deuterium atom will be retained in the final product.

(Choice D) This structure depicts the α,β-unsaturated carbonyl product containing a deuterium at the alpha position. Based on the mechanism of an aldol condensation, a deuterated aldehyde (Compound **X**) will form a product containing a deuterium at the beta position.

Educational objective:
The mechanism of an aldol condensation can be used to predict the structure of final products if an alternative reactant is used. Isotopically labeled reactants may or may not generate isotopically labeled products, depending on the position of isotope incorporation.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403813
System:
Functional Groups and Their Reactions

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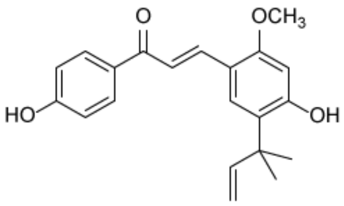

Licochalcone A

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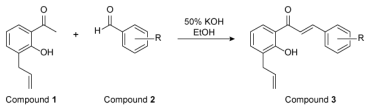

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Based on Table 1, which of the following statements does NOT correctly describe a conclusion that can be drawn about the melting point of prepared chalcone derivatives?

- A. Halogenated products have a lower melting point than ether-containing products. [9%]
- B. Compounds with hydrogen bond donors do not have the highest melting points. [22%]
- ✔ C. An increase in alkyl groups increases the melting point of the product. [54%]
- D. Nothing can be inferred about the relationship of electron-withdrawing groups on the melting point. [13%]

Correct
55% answered correctly

Explanation:

Effects of alkyl groups on melting point

Alkylated version of **3d**

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The **melting point** of a substance is the temperature at which a sample undergoes a solid-to-liquid phase change. Melting point ranges are influenced by several factors related to intermolecular forces; more and stronger intermolecular forces are present in compounds with higher melting points.

Among organic compounds with the same functional group, melting points are generally proportional to molecular weight and/or total surface area (eg, impact of branching). Larger molecules contain more atoms, more electrons, and a greater number of attractive London dispersion forces. Comparisons between functional groups require evaluation of the intermolecular forces present. Molecules that contain polar bonds demonstrate attractive dipole-dipole forces. A hydrogen bond is a particularly strong type of dipole-dipole interaction between a **hydrogen bond donor** and a **hydrogen bond acceptor**.

In Table 1, the six chalcone derivatives all have similar molecular formulae and thus similar molecular weights. Consequently, comparisons are required between the structure of the R group (column 2) and the melting point (column 5) to draw conclusions about their melting point:

- Halogenated Compound **3e** has a lower melting point than ether-containing Compound **3a (Choice A)**.
- The compounds with alcohol-containing R groups (Compounds **3b**-**3d**), which are the only derivatives that contain hydrogen bond donors, do not have the highest melting point (Compound **3f**) **(Choice B)**.
- Between Compound **3a** and Compound **3d**, the only pair of compounds that differ by an alkyl group, the added alkyl group in Compound **3a** *decreases* the melting point relative to Compound **3d**.
- Compared to the rest of the panel, the compounds containing strong electron-withdrawing groups (Compound **3e** and Compound **3f**) have the lowest and highest melting points, respectively. Therefore, it is not possible to identify a relationship between electron-withdrawing character and melting point **(Choice D)**.

The question asks which statement does NOT describe a conclusion that can be drawn about the melting point of the compounds in Table 1. Compounds **3a** and **3d** differ by an alkyl group, and Table 1 shows that the added alkyl group in Compound **3a** *decreases* the melting point relative to Compound **3d**.

Educational objective:
The melting point of an organic compound is proportional to its contributing intermolecular forces, including London dispersion forces, dipole-dipole forces, hydrogen bonding, and ionic forces. Intermolecular forces can be inferred from molecular structure and the presence of functional groups.

Time spent:0 sec

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Subject:
Organic Chemistry

System:
Functional Groups and Their Reactions